

Generation Interconnection System Impact Study Report

For Queue Project Q20-1407

Albreda Tans Mtn Tap 138 kV

May 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1407. The Interconnection Customer proposes a 235.7 MW solar photovoltaic facility interconnecting at the Albreda Tans Mtn Tap 138 kV substation in the Virginia area. The request was received on February 14, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 482 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$37,262,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Proffit Substation – Gordonsville 138 kV	Fluvanna P.S Substation – Louisa Generation Facility 345 kV	165.9	191	57.0
Colgate – Proffit 138 kV	Fluvanna P.S – Louisa Generation Facility 345 kV Line	165.6	202	57.0
Colgate – Albreda Tans Mtn Tap 138 kV	Louisa Generation Facility – Fluvanna P.S 345 kV	163.4	228	57.0
Pine Hollow – Charlottesville 138 kV line	Colgate – Albreda Tans Mtn Tap Substation 138 kV	107.9	243	100.0
Albreda Tans Mtn Tap Substation – Running Y 138 kV	Colgate – Albreda Tans Mtn Tap 138 kV	105.8	231	100.0
Nauvoo – Possum Pt 345 kV	Birchwood – Nauvoo 345 kV line	104.0	1,236	6.1
Red Hill 345 kV/138 kV Transformer	Clover – Bear 345 kV	103.7	506	8.7
Thomaston – Helijet 345 kV	Thomaston – Comanche Peak 345 kV	101.2	1,328	6.4
Peninsula – Running Y Substation 138 kV Line	Colgate – Albreda Tans Mtn Tap 138 kV Line	100.4	235	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Proffit Substation – Gordonsville 138 kV	Rebuild line with higher-capacity conductor	\$9,165,000	90% (\$8,267,000)
Colgate – Proffit 138 kV Line	Rebuild line with higher-capacity conductor	\$6,026,000	90% (\$5,436,000)
Colgate – Albreda Tans Mtn Tap 138 kV Line	Rebuild line with higher-capacity conductor	\$5,328,000	90% (\$4,806,000)
Pine Hollow – Charlottesville 138 kV line	Reconductor line	\$4,114,000	85% (\$3,499,000)
Albreda Tans Mtn Tap Substation – Running Y 138 kV	Reconductor line	\$1,878,000	100% (\$1,878,000)
Nauvoo – Possum Pt 345 kV	Reconductor line	\$22,914,000	3% (\$618,000)
Red Hill 345 kV/138 kV TX	Replace transformer with higher-rated unit	\$28,033,000	29% (\$8,178,000)
Thomaston – Helijet 345 kV	Reconductor line	\$15,474,000	5% (\$837,000)
Peninsula – Running Y Substation 138 kV Line	Reconductor line	\$3,743,000	100% (\$3,743,000)
	Total allocated to project		\$37,262,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.